Lecture 03

How to Build a Game World

Modern Game Engine - Theory and Practice

WANG XI

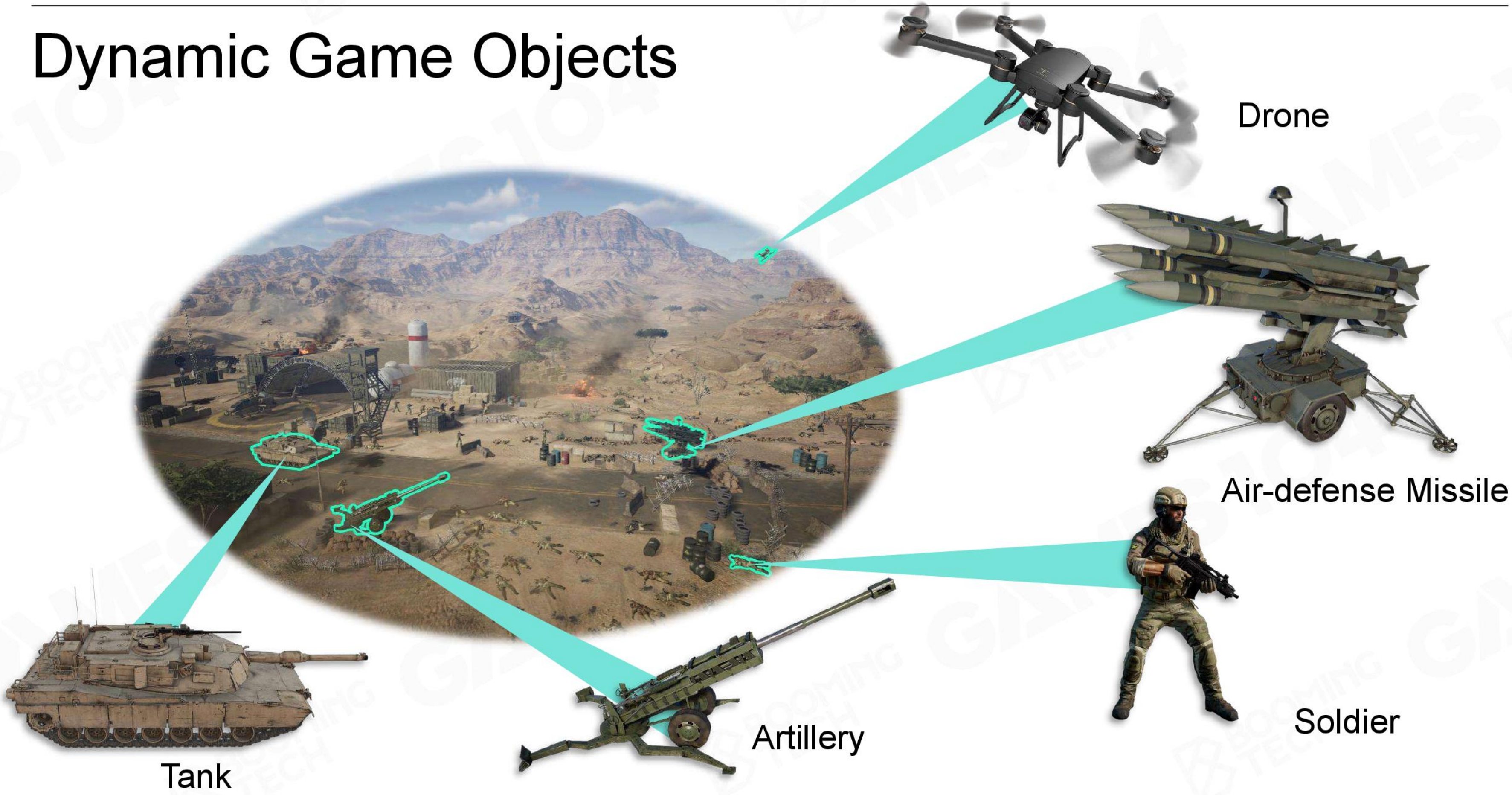
GAMES 104

2022

- 3rd Party Libraries
- Tool Layer
- Function Layer
- Resource Layer
- Core Layer
- Platform Layer

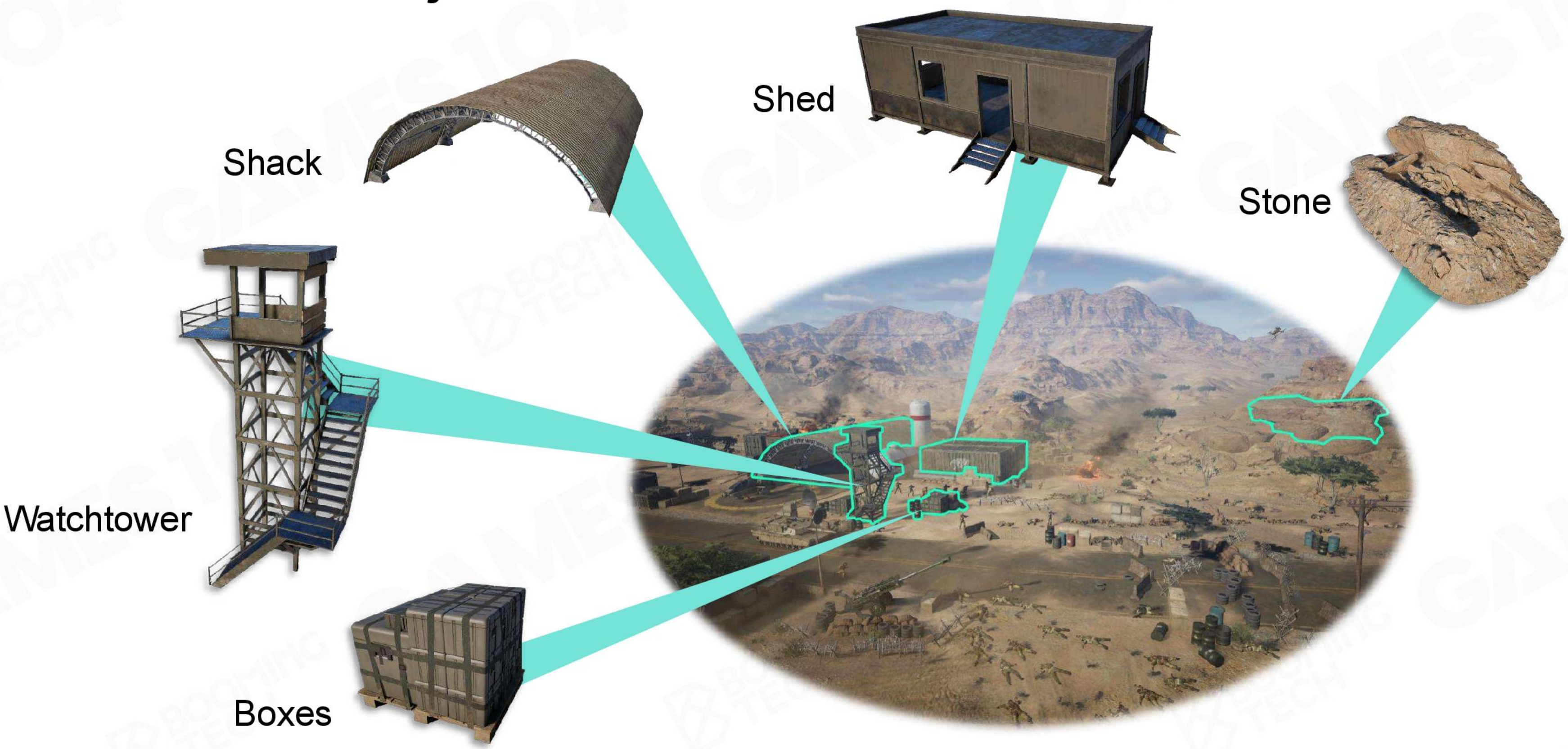
- What does a game world consist of?
- How should we describe these things?
- How are these things organized?

Dynamic Game Objects



시간의 흐름에 따라, 사용자의 입력에 따라 상태가 바뀌는 객체들

Static Game Objects

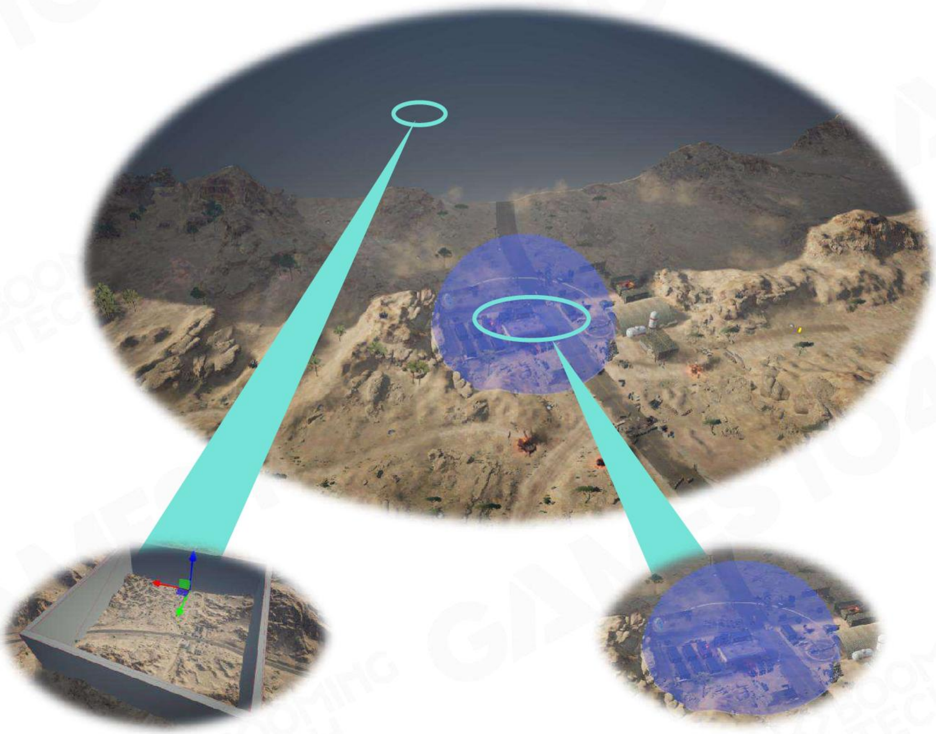


시간이 흘러도 변화가 없는 객체들

Environments

게임 공간을 만들어내는 용도로만 사용

Other Game Objects



Air wall

Trigger Area

```
function ClientRulerBase:tick(delta_time)

    local os_utility = g_context.m_os_utility;
    local current_time_stamp = os_utility:getMil

    local current_level = g_context.m_level_mana
    local game_scene = current_level.m_scene;
    local scene_loading_status = game_scene:getS

    self:tickKickAFK(delta_time);
    self:tickBoss(delta_time);

    if self.m_game_status == ClientGameStatus._I

        local ruler_type_name = self.m_definitic
```

Ruler

Navigation mesh

게임 플레이를 제한하거나 장면의 전환 등 게임 제어/제약 요소들

Everything is a Game Object

Game Object (GO)

How to Describe a Game Object?

I want a drone!

How Do We Describe a Drone in Reality?

- Shape (property)
- Position (property)
- Move (behavior)
- Capacity of battery (property)
- Etc.

객체지향적 관점: 객체는 클래스의 구현
- 같은 종류의 특성과 행동을 가짐
- 가지는 특성은 각각의 객체마다 달라짐

Game Object

Name	Drone
Property	position
	health
	battery
Behavior	move
	scout


```
class Drone
{
public:
    /* Properties */
    Vector3 position;
    float health;
    float fuel;
    ...
    /* Bebavior */
    void move();
    void scout();
    ...
};
```

Drone vs. Armed Drone

```cpp
class Drone
{
public:
    /* Properties */
    Vector3 position;
    float health;
    float fuel;
    ...
    /* Bebavior */
    void move();
    void scout();
    ...
};
```

Name	Drone	ArmedDrone
Property	position	position
	health	health
	battery	battery
		ammo
Behavior	move	move
	scout	scout
		fire

```cpp
class ArmedDrone
{
public:
    /* Properties */
    Vector3 position;
    float health;
    float fuel;
    float ammo;
    ...
    /* Bebavior */
    void move();
    void scout();
    void fire();
    ...
}
```

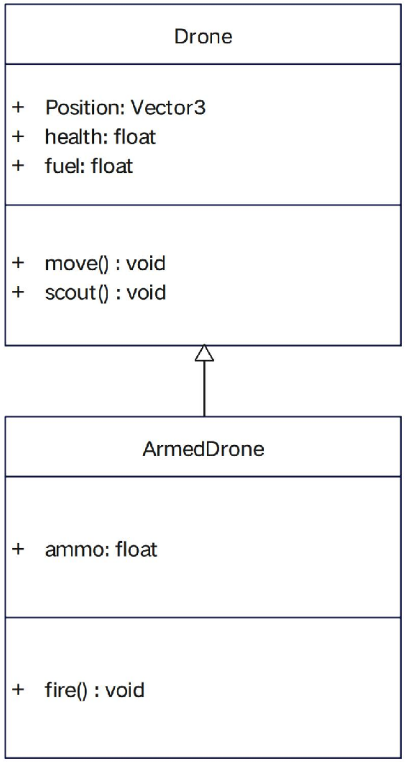
Game Object

- Inheritance

```
class Drone
{
public:
    /* Properties */
    Vector3 position;
    float health;
    float fuel;
    ...
    /* Bebavior */
    void move();
    void scout();
    ...
};
```

```
class ArmedDrone:
    public Drone
{
public:
    float ammo;
    void fire();

};
```

Code



UML Class Diagram

No Perfect Classification in the Game World!

Component Base

- Component Composition in the Real World

Excavator

Loader

Road Roller

Excavator

Bulldozer

게임 객체는 컴포넌트의 조립체

Components of a Drone

Component

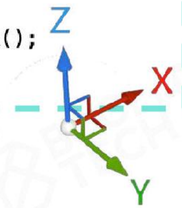
- Code example

Base class of component

```
class ComponentBase
{
    virtual void tick() = 0;
    ...
};
```

```
class GameObject
{
    vector<ComponentBase*> components;
    virtual void tick();
    ...
};
```

```
class TransformComponent:
    public ComponentBase
{
    Vector3 position;
    ...
    void tick();
};
```




```
class ModelComponent:
    public ComponentBase
{
    Mesh mesh;
    ...
    void tick();
};
```

```
class MotorComponent:
    public ComponentBase
{
    float battery;
    void tick();
    void move();
    ...
};
```

```
class AIComponent:
    public ComponentBase
{
    void tick();
    void scout();
    ...
};
```

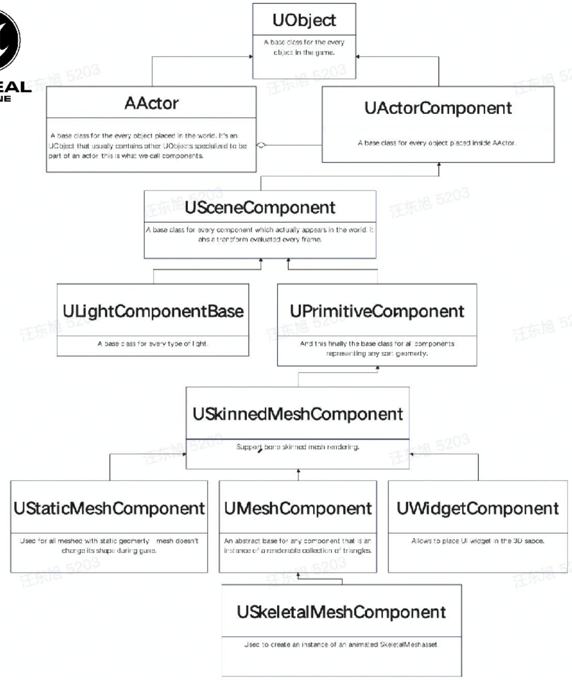
Animation
Physics
...

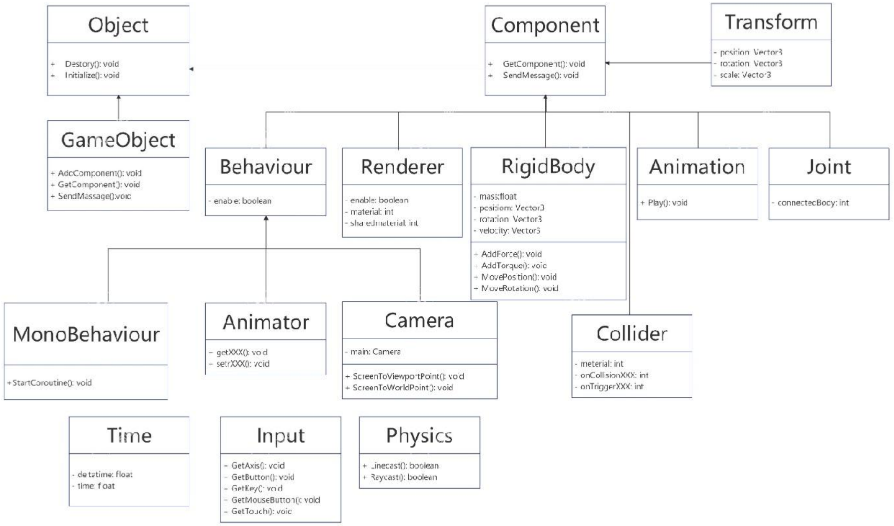
Component

- Drone vs. Armed Drone

Components		Components
Transform	=	Transform
Model	=	Model
Animation	=	Animation
Motor	=	Motor
Physics	=	Physics
…	=	…
AI	≠	AI
		Combat

Components in Commercial Engines





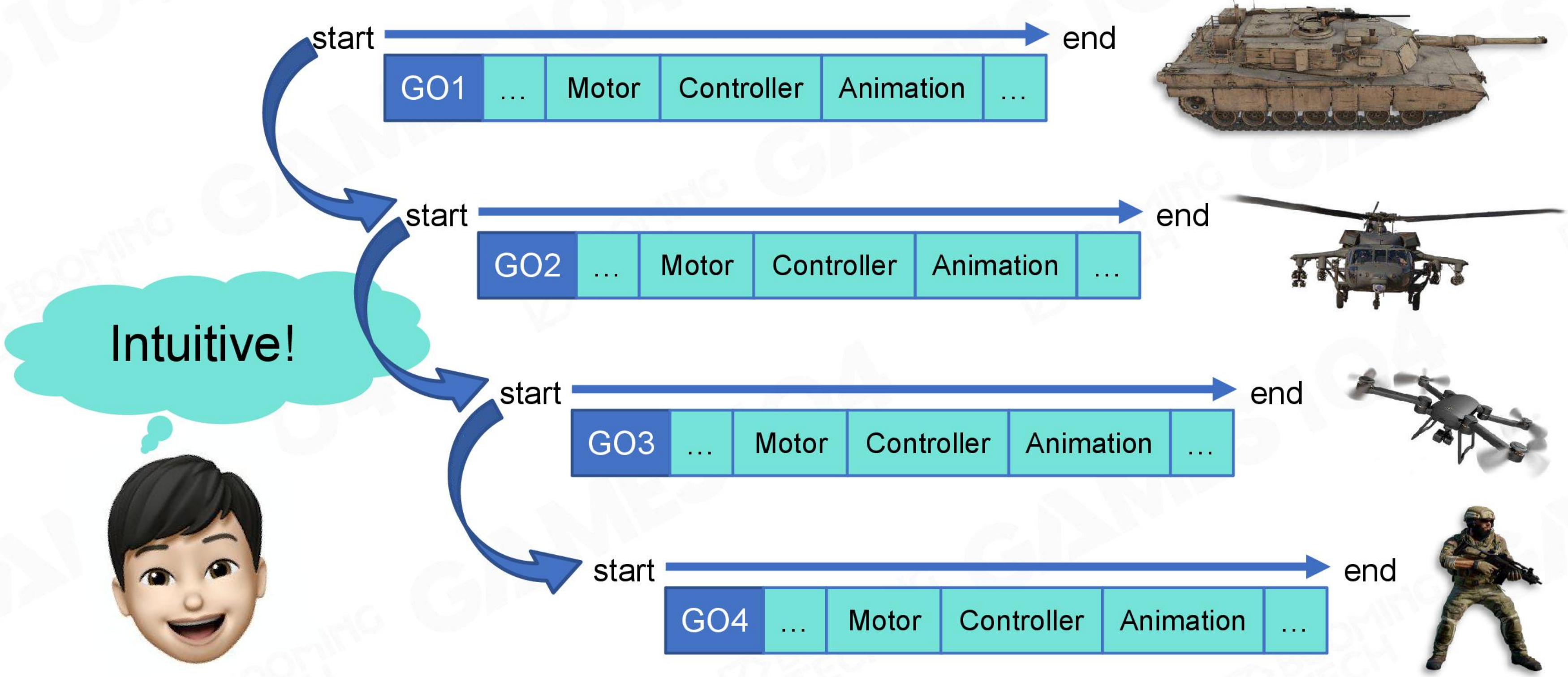
For a more detailed version: https://app.creately.com/d/fZknuArLuyk/view

Takeaways

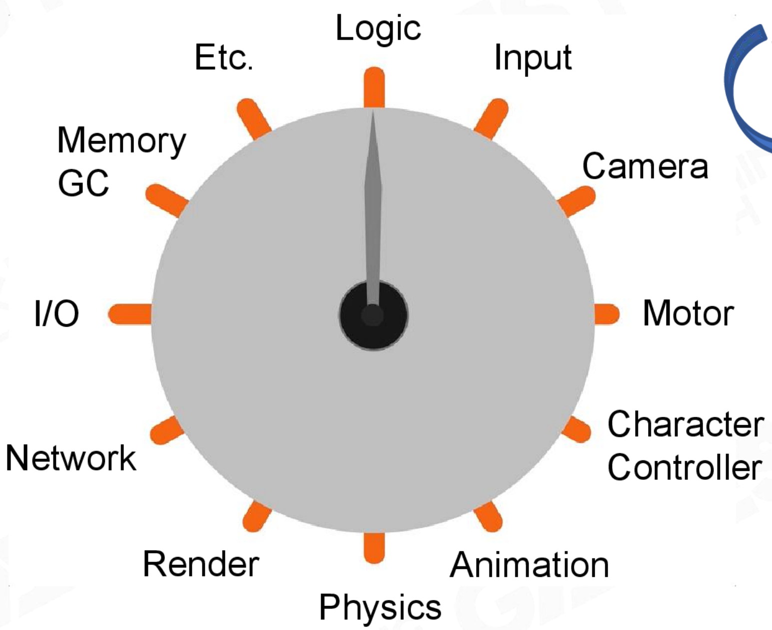
- Everything is a game object in the game world
- Game object could be described in the component-based way

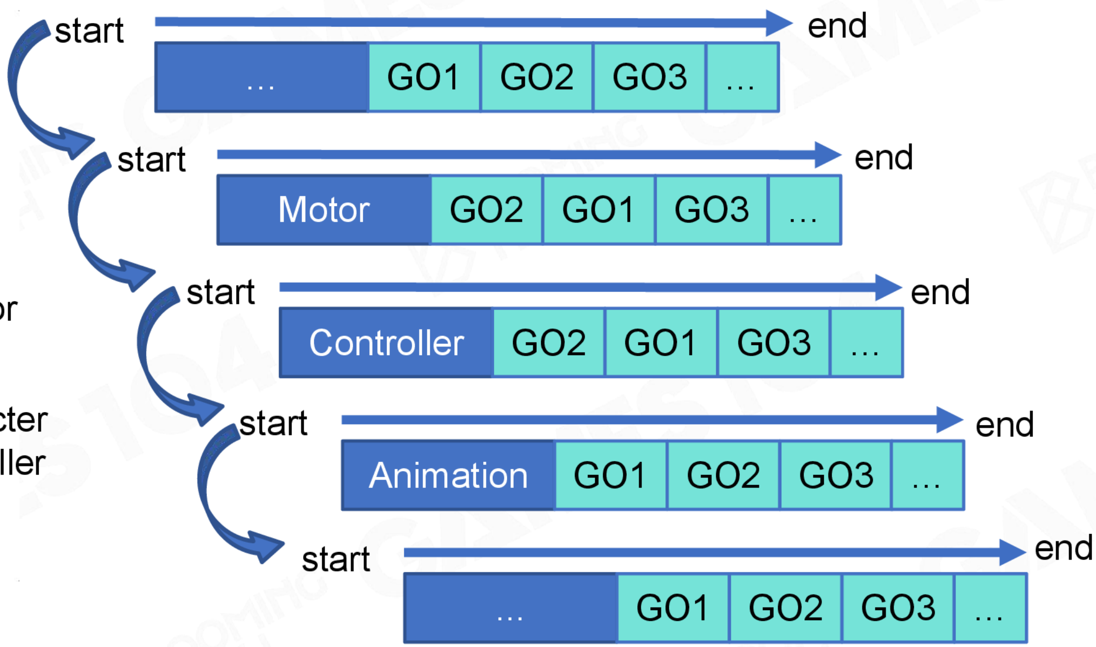
How to Make the World Alive?

Object-based Tick



Component-based Tick

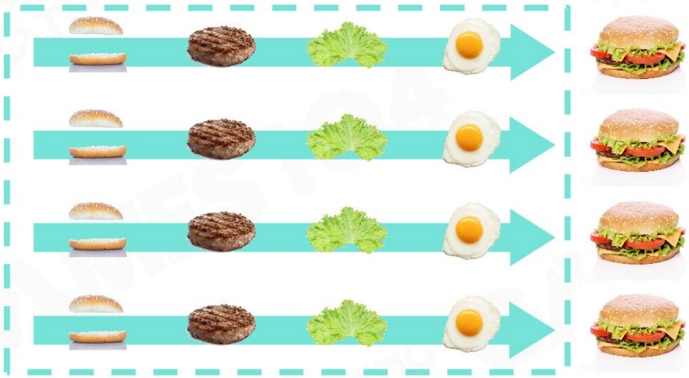


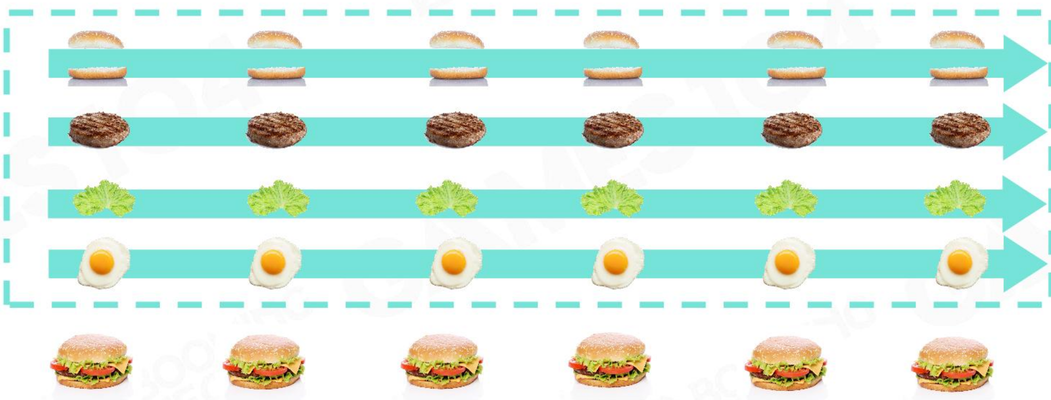


Object-based Tick vs. Component-based Tick

More efficient!

- Object-based tick
  - Simple and intuitive
  - Easy to debug
- Component-based tick
  - Parallelized processing
  - Reduced cache miss





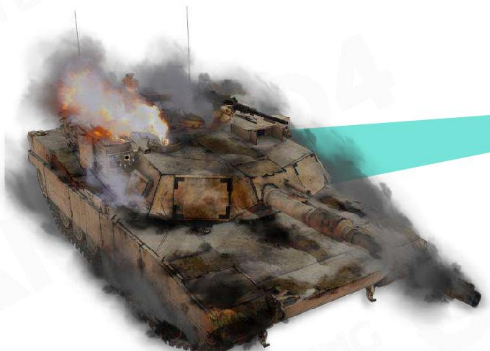
How to Explode an Ammo in a Game?

이벤트의 전파와 처리

Hardcode

Soldier

Helicopter



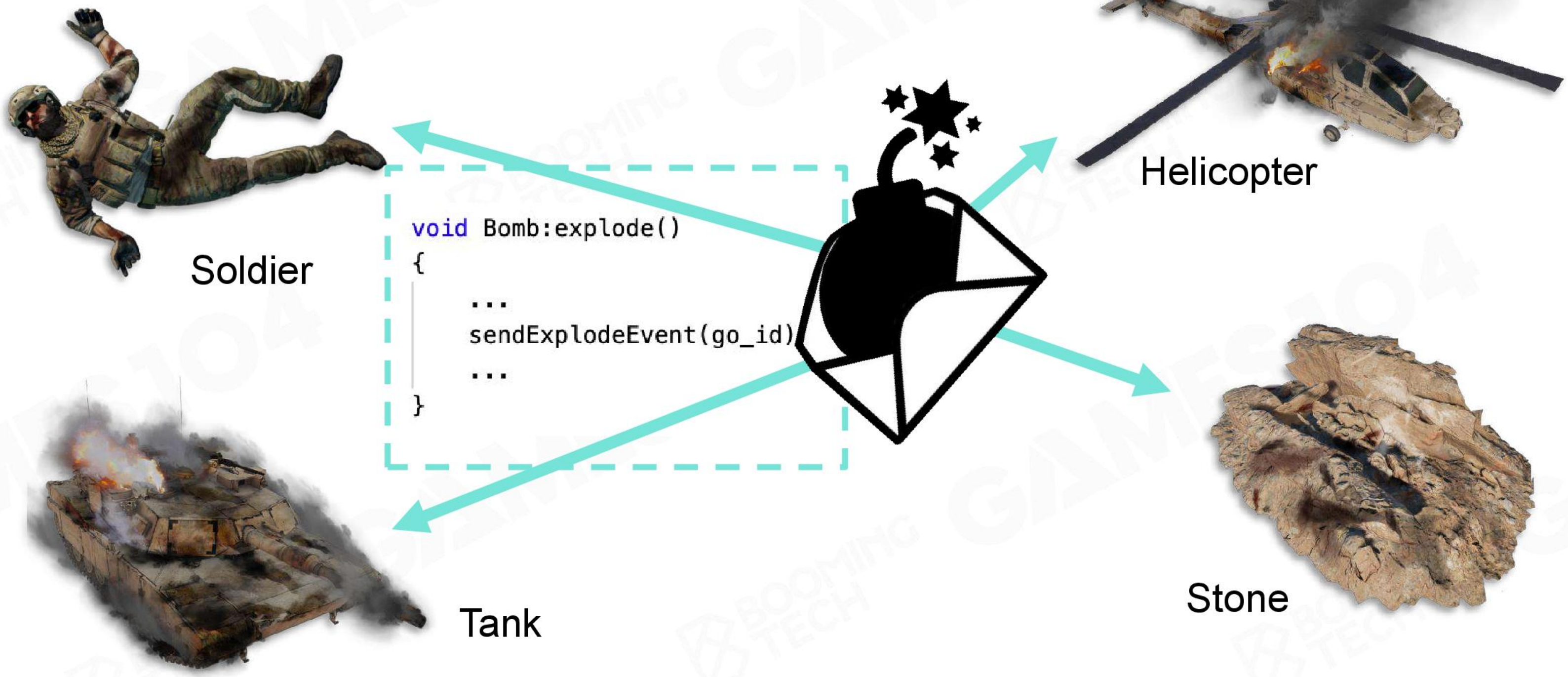
Tank

Stone

```
void Bomb:explode()
{
    ...
    switch(go_type)
    {
        case GoType.humen_type:
        {
            /* process soldier */
        }
        case GoType.drone_type:
        {
            /* process drone */
            ...
        }
        case GoType.tank_type:
        {
            /* process tank */
            ...
        }
        case GoType.stone_type:
        {
            /* process stone */
            ...
        }
        default:
        {
            break;
        }
    }
}
```

Events

- Message sending and handling
- Decoupling event sending and handling



Events Mechanism in Commercial Engines

unity

```csharp
using UnityEngine;

public class Example : MonoBehaviour
{
    void Start()
    {
        // Calls the function ApplyDamage with a value of 5
        // Every script attached to the game object
        // that has an ApplyDamage function will be called.
        gameObject.SendMessage("ApplyDamage", 5.0);
    }
}

public class Example2 : MonoBehaviour
{
    public void ApplyDamage(float damage)
    {
        print(damage);
    }
}
```

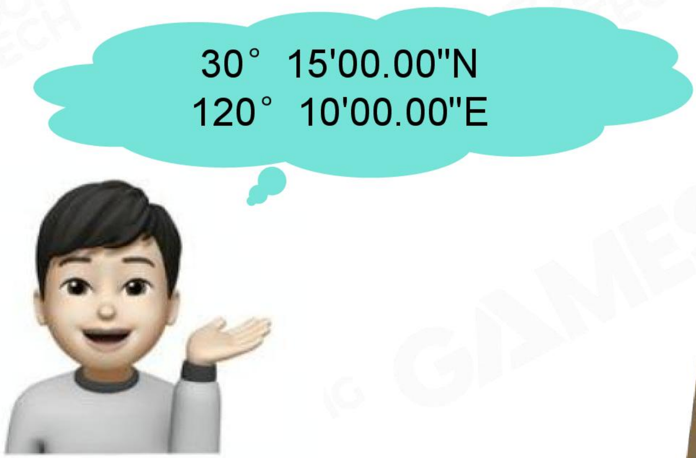
UNREAL ENGINE

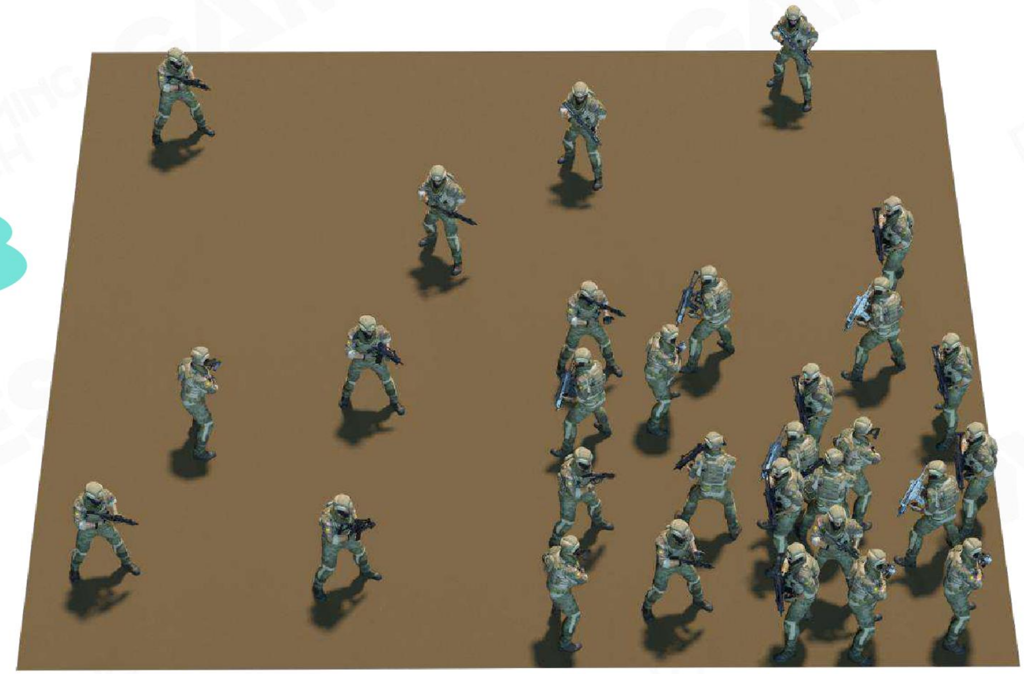
```cpp
/** Event for when collections are created */
DECLARE_EVENT_OneParam( ICollectionManager, FCollectionCreatedEvent, const FCollectionNameType& );
virtual FCollectionCreatedEvent& OnCollectionCreated() = 0;

/** Event for when collections are destroyed */
DECLARE_EVENT_OneParam( ICollectionManager, FCollectionDestroyedEvent, const FCollectionNameType& );
virtual FCollectionDestroyedEvent& OnCollectionDestroyed() = 0;
    Tim Sweeney, 8年前 • Engine source (Main branch up to CL 2026164) …
/** Event for when assets are added to a collection */
DECLARE_EVENT_TwoParams( ICollectionManager, FAssetsAddedEvent, const FCollectionNameType&, const TArray<FName>& );
virtual FAssetsAddedEvent& OnAssetsAdded() = 0;

/** Event for when assets are removed from a collection */
DECLARE_EVENT_TwoParams( ICollectionManager, FAssetsRemovedEvent, const FCollectionNameType&, const TArray<FName>& )
virtual FAssetsRemovedEvent& OnAssetsRemoved() = 0;

/** Event for when collections are renamed */
DECLARE_EVENT_TwoParams( ICollectionManager, FCollectionRenamedEvent, const FCollectionNameType&, const FCollection
virtual FCollectionRenamedEvent& OnCollectionRenamed() = 0;

/** Event for when collections are re-parented (params: Collection, OldParent, NewParent) */
DECLARE_EVENT_ThreeParams( ICollectionManager, FCollectionReparentedEvent, const FCollectionNameType&, const TOptior
virtual FCollectionReparentedEvent& OnCollectionReparented() = 0;

/** Event for when collections is updated, or otherwise changed and we can't tell exactly how (eg, after updating fr
DECLARE_EVENT_OneParam( ICollectionManager, FCollectionUpdatedEvent, const FCollectionNameType& );
virtual FCollectionUpdatedEvent& OnCollectionUpdated() = 0;

/** When a collection checkin happens, use this event to add additional text to the changelist description */
DECLARE_EVENT_TwoParams( ICollectionManager, FAddToCollectionCheckinDescriptionEvent, const FName& /*CollectionName
virtual FAddToCollectionCheckinDescriptionEvent& OnAddToCollectionCheckinDescriptionEvent() = 0;
;
```

How to Manage Game Objects?

Scene Management

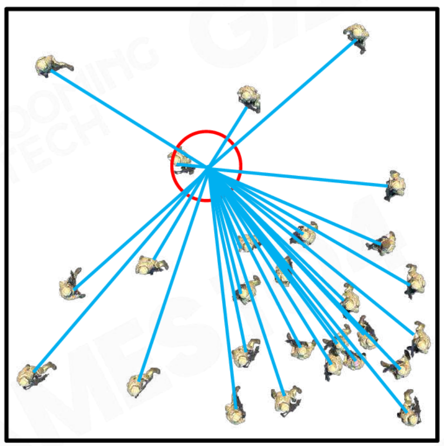
- Game objects are managed in a scene
- Game object query
  - By unique game object ID
  - By object position



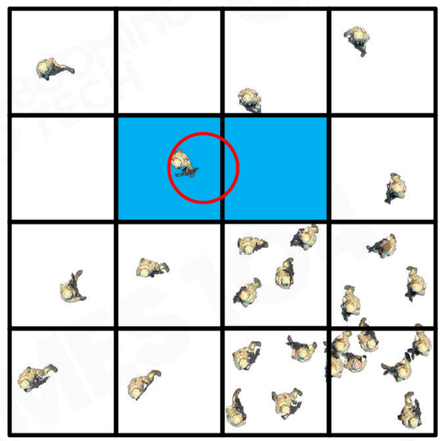


Scene Management

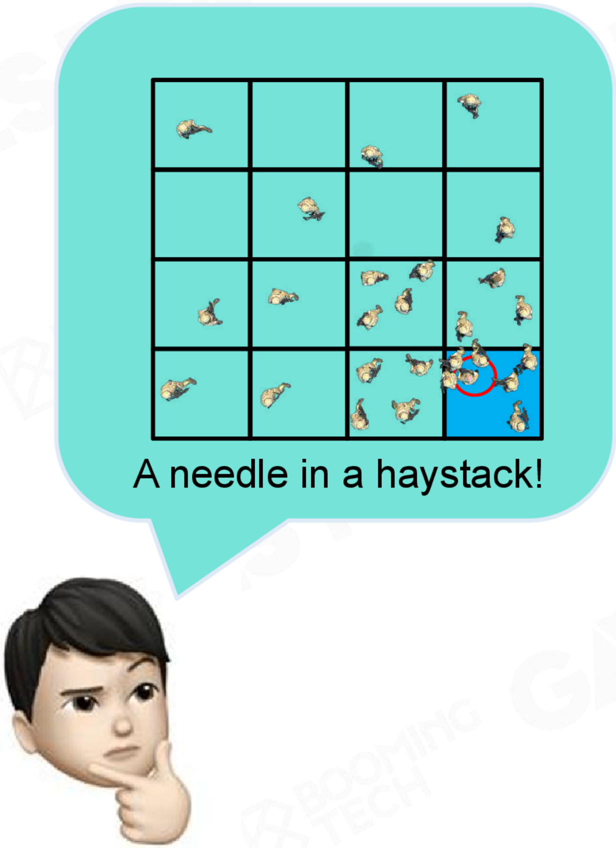
- Simple space segmentation



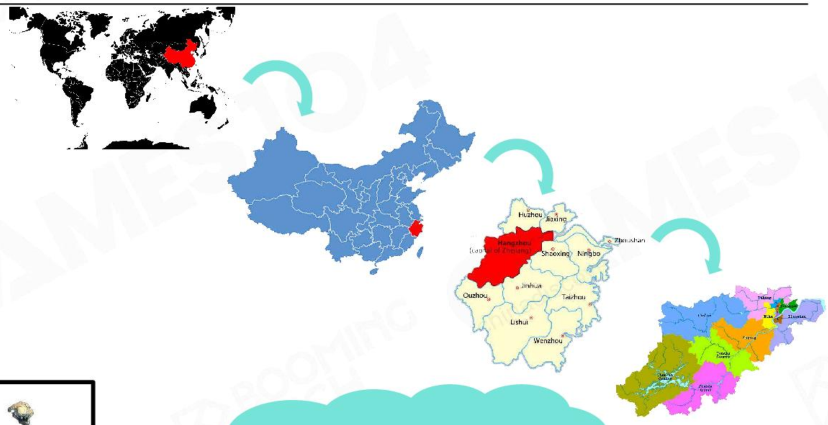
No division



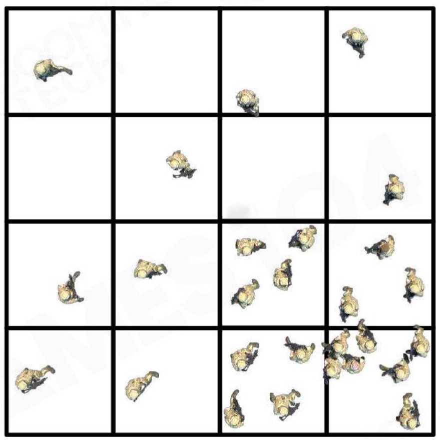
Divided by grid



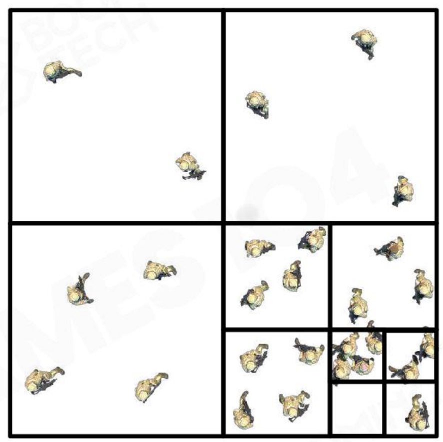
Scene Management



- Segmented space by object clusters
- Hierarchical segmentation



Divided by grid

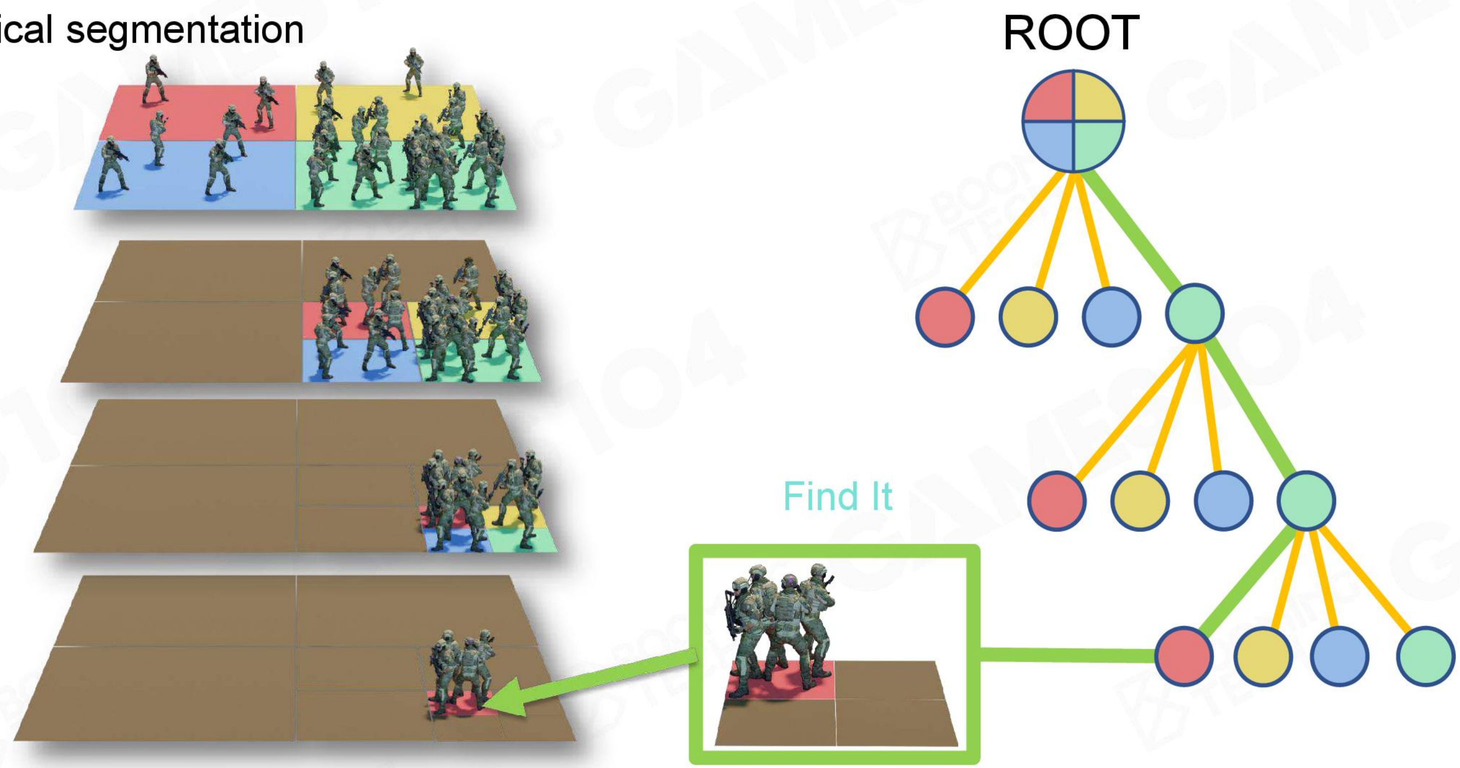


Quadtree

Scene Management

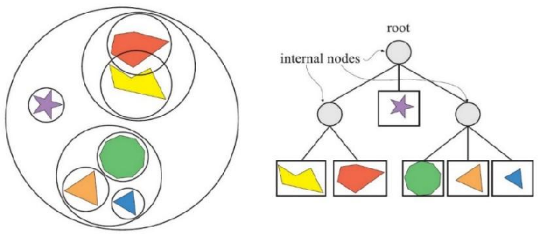
- Segmented space by object clusters
- Hierarchical segmentation

충돌 감지를 상상해 보라

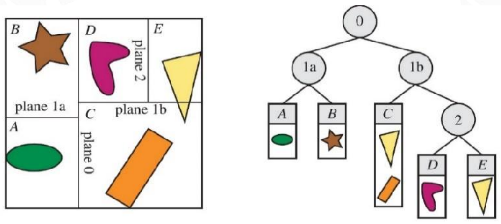


Scene Management

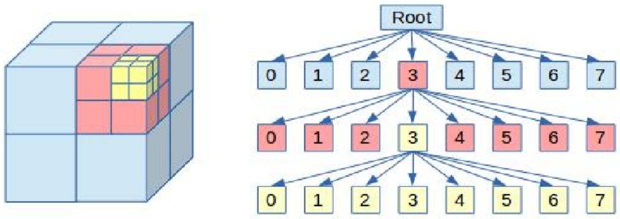
- Spatial Data Structures



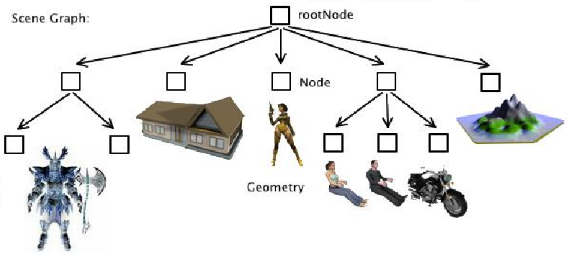
Bounding Volume Hierarchies (BVH)



Binary Space Partitioning(BSP)



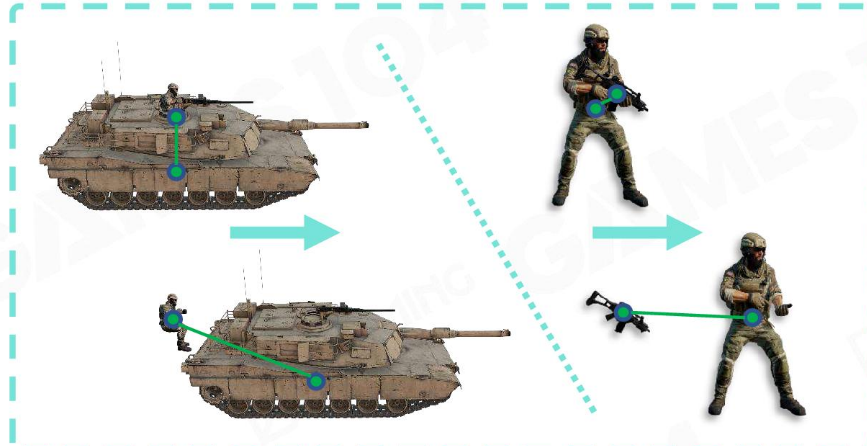
Octree



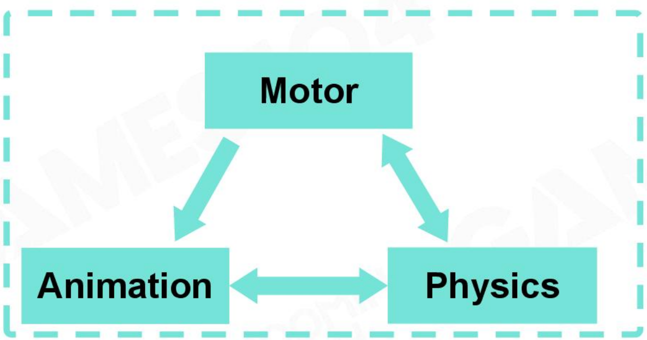
Scene Graph

Takeaways

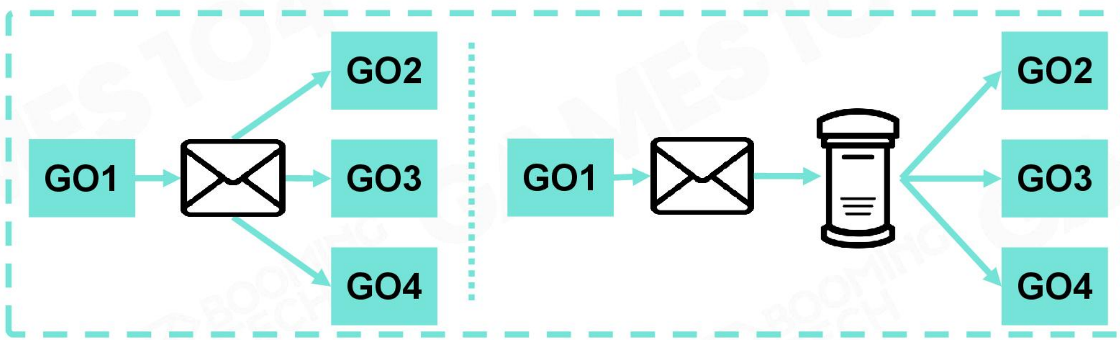
- Everything is an object
- Game object could be described in the component-based way
- States of game objects are updated in tick loops
- Game objects interact with each other via event mechanism
- Game objects are managed in a scene with efficient strategies



GO Bindings



Component Dependencies



Immediate Event Sending or not

Modern Game Engine - Theory and Practice

Course Survey

Please scan here to let us know what you think

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- Jason
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- BOOK
- MANDY
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- 阿乐
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Q&A

Enjoy ;) Coding

Course Wechat

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